

System Architecture & Production Design

Intelligence Graph Data Pipeline | Scalability & Resilience Strategy

1. Scale Strategy: Acquiring 500,000+ Records

To transition from a localized script to a massive, one-time bulk extraction system capable of handling hundreds of thousands of records without manual intervention, the architecture must be decoupled and distributed.

- **Distributed Crawler Nodes:** The scraping logic will be containerized via Docker and orchestrated using Kubernetes (K8s). This allows horizontal scaling of worker nodes (e.g., using Celery or AWS SQS) to process thousands of URLs concurrently.
- **URL Frontier Management:** A robust message broker (RabbitMQ or Apache Kafka) will manage the queue of target URLs. This ensures fault tolerance; if a worker node crashes, the unacknowledged URL is simply returned to the queue for another node to process.
- **Proxy Rotation:** To prevent IP bans at scale, requests will be routed through a residential proxy network with automatic IP rotation and geographic targeting.

2. Handling Context Windows (413) & Rate Limits (429)

LLMs are the bottleneck in modern data pipelines. Managing their constraints requires strict programmatic safeguards.

- **Managing 413s (Payload Too Large):** Before routing raw HTML to the LLM, the system runs an intelligent DOM stripping function (removing `